MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE**
Course Name	**Construction Materials and Quality Management**	**Course Code**	**CIE 4053**
Semester	**VII**	**Max Marks**	**50**
Date	**29/12/2024**	**Duration**	**3 Hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

<table>
<tr><th rowspan="2">Q. No.</th><th rowspan="2">Question</th><th rowspan="2">Marks</th><th colspan="2">NBA</th><th rowspan="2">BL</th></tr>
<tr><th>CO</th><th>PO</th></tr>
<tr><td>1A</td><td>Interpret the total savings possible using hind-sight strategy over conservative strategy for the following data. Take monthly requirement of 50 units and optimal storage capacity of 2 months' requirement.
<table>
<tr><th>Month</th><th>Price (INR)</th><th>Month</th><th>Price(INR)</th></tr>
<tr><td>1</td><td>682</td><td>8</td><td>692</td></tr>
<tr><td>2</td><td>683</td><td>9</td><td>715</td></tr>
<tr><td>3</td><td>679</td><td>10</td><td>728</td></tr>
<tr><td>4</td><td>719</td><td>11</td><td>685</td></tr>
<tr><td>5</td><td>712</td><td>12</td><td>676</td></tr>
<tr><td>6</td><td>710</td><td>13</td><td>704</td></tr>
<tr><td>7</td><td>704</td><td>14</td><td>704</td></tr>
</table></td><td>4</td><td>3</td><td>5,7</td><td>2</td></tr>
<tr><td>1B</td><td>a) Explain the significance of i) Right quality
ii) Right price in Purchase Management.
b) Explain the necessity of price forecasting.</td><td>3</td><td>3</td><td>2,3</td><td>3</td></tr>
<tr><td>1C</td><td>Describe the significance and objectives of material planning. For a construction project of your choice, prepare a comprehensive WBS.</td><td>3</td><td>2</td><td>1,11</td><td>2</td></tr>
</table>

<table>
<tr><td>2 A</td><td>Explain with an example each, the Alpha-numeric and Kodak systems of codification.</td><td>3</td><td>2</td><td>3,5,7</td><td>3</td></tr>
<tr><td>2 B</td><td>Compare the effectiveness of price forecasting strategies between weighted moving average and average methods. Take (N=3, w_1=0.25, w_2=0.35, w_3=0.4).
<table>
<tr><th>Month</th><th>Price for Year- 1 (INR)</th><th>Price for Year-2 (INR)</th></tr>
<tr><td>January</td><td>135</td><td>141</td></tr>
<tr><td>February</td><td>140</td><td>129</td></tr>
<tr><td>March</td><td>117</td><td>117</td></tr>
<tr><td>April</td><td>159</td><td>138</td></tr>
<tr><td>May</td><td>131</td><td>131</td></tr>
<tr><td>June</td><td>133</td><td>151</td></tr>
<tr><td>July</td><td>157</td><td>142</td></tr>
<tr><td>August</td><td>151</td><td>154</td></tr>
<tr><td>September</td><td>140</td><td>144</td></tr>
<tr><td>October</td><td>125</td><td>125</td></tr>
<tr><td>November</td><td>159</td><td>131</td></tr>
<tr><td>December</td><td>144</td><td>151</td></tr>
</table></td><td>5</td><td>3</td><td>5,7</td><td>3</td></tr>
<tr><td>2 C</td><td>What is house of quality in quality management? Explain with an example.</td><td>2</td><td>5</td><td>1,2</td><td>2</td></tr>
<tr><td>3 A</td><td>Execute XYZ analysis on the data tabulated below. Take ordering cost as 20 % and carrying Cost as 25 % of unit cost of material item.</td><td>5</td><td>2</td><td>3,5,7</td><td>3</td></tr>
</table>

<table>
<tr><td></td><td>
<table>
<tr><th>Item. No</th><th>Unit Price (×100)(₹)</th><th>No of units held annually (×100)</th></tr>
<tr><td>1</td><td>11</td><td>160</td></tr>
<tr><td>2</td><td>1</td><td>200</td></tr>
<tr><td>3</td><td>30</td><td>90</td></tr>
<tr><td>4</td><td>518</td><td>120</td></tr>
<tr><td>5</td><td>135</td><td>180</td></tr>
<tr><td>6</td><td>2</td><td>10</td></tr>
<tr><td>7</td><td>600</td><td>140</td></tr>
<tr><td>8</td><td>420</td><td>300</td></tr>
<tr><td>9</td><td>45</td><td>90</td></tr>
<tr><td>10</td><td>2</td><td>15</td></tr>
</table>
</td><td></td><td></td><td></td><td></td></tr>
<tr><td>3 B</td><td>Discuss the factors that govern the strategy for purchase of capital equipment. Bring out the essential differences between the thumb rule and use of time series data for price forecasting.</td><td>3</td><td>5</td><td>1,2</td><td>2</td></tr>
<tr><td>3 C</td><td>“Judicious selection of quantity to be purchased at periodic interval such that always purchase is effected at lowest possible price within limitations imposed by constraints” Justify whether the above strategy follows the Integrated approach.</td><td>2</td><td>1</td><td>1,2</td><td>2</td></tr>
<tr><td>4 A</td><td>Discuss all the methods of selective control and Explain any 2.</td><td>5</td><td>2</td><td>5,7</td><td>3</td></tr>
<tr><td>4 B</td><td>For the following data of capital investment ₹ 1,00,000 and expected salvage value ₹ 10,000, suggest if the investment is justified. The alternate investment consists of 15 % return within 5 years
<table>
<tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>Return after deduction(INR)</td><td>30000</td><td>28000</td><td>24000</td><td>18000</td><td>16000</td></tr>
</table>
</td><td>3</td><td>5</td><td>5,6</td><td>2</td></tr>
<tr><td>4 C</td><td>Explain pareto chart and control chart.</td><td>2</td><td>5</td><td>5,6</td><td>2</td></tr>
<tr><td>5 A</td><td>Explain the methods to generate preferred numbers for standardization of the products.</td><td>4</td><td>2</td><td>5,7</td><td>3</td></tr>
<tr><td>5 B</td><td>a) Explain about moving average and exponential smoothening methods.</td><td>4</td><td>3</td><td>3</td><td>3</td></tr>
</table>

	b) Discuss the **limitations** of moving average method of price forecasting. Explain how they can be **overcome** by weighted moving average and exponential smoothening methods?				
5 C	Describe any FOUR benefits of price forecasting strategy of material purchase.	**2**	3	5,7	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***2***
2	***20***
3	***18***
4	***0***
5	***10***

BL	Total Marks
2	**30**
3	**20**
4	**0**
5	**0**